Marton's Inner Bound for Covert Communications

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***Abstract*—We introduce the covert Marton inner bound for a discrete memoryless broadcast channel with an external warden, and present a simplified expression for this bound. In particular, we show that a certain parametrization of the auxiliary random variables (U, V, T) is optimal for the evaluation of the covert Marton region.**

I. Introduction and Channel Model

Covert communication considers communication systems in which an external adversary, referred to as the warden, should not be able to detect whether communication is taking place. This requirement is typically imposed by constraining the Kullback-Leibler (KL) divergence between the warden's output distribution induced by communication and the output distribution induced when the transmitter sends the innocent symbol x_0 at every channel use. Under standard non-redundancy conditions, this constraint leads to the square-root law: the number of reliably communicated information bits scales as the square root of the blocklength. Covert capacities for point-to-point discrete memoryless and Gaussian channels were characterized in [1], [2], [3], [4], [5].

Covert communication over broadcast channels (BC) was studied in [6], [7], [8]. The works in [6], [7] considered mixed covert/non-covert communication, whereas [8] considered the setting where the transmitter sends individual messages to two receivers while the entire communication must remain undetectable to an external warden. In this latter setting, [8] characterized conditions under which time-sharing between the two receivers is optimal and established its optimality for several important classes of broadcast channels.

More recently, the authors [9] derived the covert capacity region of degraded BCs. It is achieved by superposition coding, and time-sharing is in general strictly suboptimal. The optimal parametrization of the superposition coding region for covert communication assigns a nonvanishing probability to one of the cloud-center symbols u_0, conditioned on which all nonzero inputs have vanishing probabilities, while the remaining cloud-center values u have vanishing probabilities and, conditioned on them, all inputs can have positive probabilities [9].

In the present work, we consider a general BC and develop a covert counterpart of Marton's region based on the one-shot Marton achievability result of [10]. We prove that an optimal parametrization of this Marton region comprises two cloud-center symbols u_1 and u_2 with nonvanishing probabilities,[1] and conditioned on u_1 we only have a V-satellite which moreover coincides with the input X, while conditioned on u_2 we only have a T-satellite again equal to X. All other cloud-center symbols have vanishing probabilities and conditioned on them we can use an arbitrary distribution on the satellites.

[1] Alternatively one can also phrase an optimal parametrization with a single u_0 of positive probability, but the two parametrizations are equivalent.

A. System Model and Definitions

Let $\mathcal{X}$, $\mathcal{Y}_1$, $\mathcal{Y}_2$, $\mathcal{Z}$ be finite sets. Denote by $\mathcal{P}(\mathcal{X})$ the class of all distributions on $\mathcal{X}$. A discrete memoryless broadcast channel (BC) with two users and a warden is a quintuple $\{\mathcal{X}, P_{Y_1Y_2Z|X}, \mathcal{Y}_1, \mathcal{Y}_2, \mathcal{Z}\}$ where $\mathcal{X}$ is the input alphabet, $\mathcal{Y}_k$ is the output alphabet of user k, for $k = 1, 2$, $\mathcal{Z}$ is the output alphabet at the warden and $P_{Y_1Y_2Z|X}$ is a transition probability matrix from $\mathcal{X}$ to $\mathcal{Y}_1 \times \mathcal{Y}_2 \times \mathcal{Z}$. We denote by $P_{Z|X}$ and $P_{Y_k|X}$, for $k = 1, 2$, the conditional marginals of $P_{Y_1Y_2Z|X}$, and by $P_X \circ P_{Z|X}$ the distribution on $\mathcal{Z}$ induced by P_X at the input. When P_X is understood from the context, we will use the notation P_Z, and similarly for $P_X \circ P_{Y_k|X}$, P_{Y_k} etc. Let $x_0 \in \mathcal{X}$ stand for the zero symbol, namely, the symbol fed into the channel when no communication is taking place. Define

$$Q_0(z) = P_{Z|X}(z|x_0), \quad z \in \mathcal{Z}, \tag{1}$$

and let Q_0 stand for the n-fold product of Q_0, i.e.,

$$Q_0^{\times n}(z^n) = \prod_{i=1}^{n} Q_0(z_i). \tag{2}$$

The goal in covert communication is to transmit information to the receivers while keeping the distribution at the warden output close to Q_0. Fix integers ν_k, $k = 1, 2$ and transmission length n. Let $\mathcal{N}_k = \{1, \ldots, \nu_k\}$ stand for the set of messages of user k. The transmitter and legitimate receivers share a random secret key S, taking values in a finite set $\mathcal{S}$. We assume that the key is of sufficiently large randomness, thus do not specify the size of $\mathcal{S}$.

Definition 1. *An $(n, \nu_1, \nu_2, \epsilon, \delta)$ covert code for the BC $\{\mathcal{X}, P_{Y_1Y_2Z|X}, \mathcal{Y}_1, \mathcal{Y}_2, \mathcal{Z}\}$ with a warden consists of an encoder*

$$f : \mathcal{N}_1 \times \mathcal{N}_2 \times \mathcal{S} \to \mathcal{X}^n, \tag{3}$$

and pair of decoders

$$\phi_k : \mathcal{Y}_k^n \times \mathcal{S} \to \mathcal{N}_k, \qquad k = 1, 2, \tag{4}$$

such that the probabilities of error are bounded by ϵ:

$$P_{e,k} = \frac{1}{\nu_1\nu_2}\sum_{\substack{(m_1,m_2)\\ \in\mathcal{N}_1\times\mathcal{N}_2}}\sum_{s\in\mathcal{S}} P_S(s)P_{Y_k^n|X^n}\left(\mathcal{D}_{m_k}^c(s)|f(m_1,m_2,s)\right) \le \epsilon, \quad k=1,2, \quad (5)$$

and the output distribution at the warden approximates $Q_0^{\times n}$ in the KL divergence sense:

$$D\left(P_{Z^n}||Q_0^{\times n}\right) \le \delta. \quad (6)$$

The set $\mathcal{D}_{m_k}(s)$ in (5) is the decoding region of message m_k

$$\mathcal{D}_{m_k}(s) = \{y_k^n : \quad \phi_k(y_k^n, s) = m_k\}, \qquad k=1,2, \quad (7)$$

and P_{Z^n} in (6) stands for the distribution of the warden output Z^n induced by the operation of the code:

$$P_{Z^n}(z^n) = \frac{1}{\nu_1\nu_2}\sum_{\substack{(m_1,m_2)\\ \in\mathcal{N}_1\times\mathcal{N}_2}}\sum_{s\in\mathcal{S}}\frac{1}{|\mathcal{S}|}P_{Z^n|X^n}(z^n|f(m_1,m_2,s)). \quad (8)$$

In this work, we focus on BCs that satisfy the following conditions, that are now standard in covert communications [4], [3], [8].

Conditions 1 (Non-redundancy and absolute continuity)**.** *a) The zero symbol is not redundant at the warden output. I.e., $Q_0 \notin CH[P_{Z|X}(\cdot|x'),\ x' \in \mathcal{X}\setminus\{x_0\}]$, where CH stands for the convex hull.*

b) Absolute continuity w.r.t. x_0 symbol at the warden: $P_{Z|X}(\cdot|x) \ll Q_0 \quad \forall x \in \mathcal{X}$.

c) Absolute continuity w.r.t. x_0 symbol at the users' outputs: $P_{Y_k|X}(\cdot|x) \ll P_{Y_k|X}(\cdot|x_0) \quad \forall x\in\mathcal{X}, \quad k=1,2$.

Part a) of Conditions 1 guarantees that the encoder cannot mimic the no communication state with an input distribution P_X that results in the warden output distribution being equal to Q_0. If part b) is not satisfied, then there is an input symbol x' that the encoder cannot use, effectively reducing the input alphabet size. For the single-user channel, it is shown in [3] that if c) does not hold, the number of covert bits that can be transmitted grows like $\sqrt{n}\log n$. Thus, Conditions 1 are the most pessimistic assumptions that still allow covert communications. For details, see [3, Appendix G] and [8].

The covert rates of the code are defined as

$$L_k = \frac{\log\nu_k}{\sqrt{n\delta}}, \quad k=1,2 \quad (9)$$

Definition 2. *A pair of covert rates (L_1, L_2) is called δ-achievable if for any $\epsilon > 0$, $\rho > 0$ and sufficiently large n there exists an $(n, 2^{\sqrt{n\delta}(L_1-\rho)}, 2^{\sqrt{n\delta}(L_2-\rho)}, \epsilon, \delta)$ covert code for the BC $P_{Y_1Y_2Z|X}$. The collection of all δ-achievable pairs is called the covert capacity region, and is denoted by $\mathcal{L}_\delta^*$.*

II. Main Results

Definition 3. *Let $\mathcal{L}_{n,\delta}^{(\mathsf{M})}$ be the set of all nonnegative pairs (L_1, L_2) that for some P_{UVT} over finite auxiliary alphabets $\mathcal{U}\times\mathcal{V}\times\mathcal{T}$ and some function $f: \mathcal{U}\times\mathcal{V}\times\mathcal{T}\to\mathcal{X}$ satisfy the following three constraints:*

$$L_1 \le \sqrt{\frac{n}{\delta}}I(U,V;Y_1), \quad (10a)$$

$$L_2 \le \sqrt{\frac{n}{\delta}}I(U,T;Y_2), \quad (10b)$$

$$L_1+L_2 \le \sqrt{\frac{n}{\delta}}\min\{I(U;Y_1), I(U;Y_2)\} + \sqrt{\frac{n}{\delta}}I(V;Y_1|U) + \sqrt{\frac{n}{\delta}}I(T;Y_2|U) - \sqrt{\frac{n}{\delta}}I(V;T|U), \quad (10c)$$

as well as the covertness constraint:

$$D\left(P_Z||Q_0\right) \le \frac{\delta}{n}. \quad (10d)$$

Definition 4. *Define the* covert Marton region

$$\mathcal{L}_\delta^{(\mathsf{M})} := \bigcap_{n\ge1}\mathcal{L}_{n,\delta}^{(\mathsf{M})}. \quad (11)$$

Theorem 1. *For each $\delta > 0$, the* covert Marton region *is δ-achievable:*

$$\mathcal{L}_\delta^{(\mathsf{M})} \subseteq \mathcal{L}_\delta^*. \quad (12)$$

In evaluating $\mathcal{L}_{n,\delta}^{(\mathsf{M})}$, it is sufficient to restrict the auxiliary alphabets to

$$|\mathcal{U}| \le |\mathcal{X}|+4, \qquad |\mathcal{V}| \le |\mathcal{X}|, \qquad |\mathcal{T}| \le |\mathcal{X}|. \quad (13)$$

This follows from the cardinality constraints in [11]. The argument there immediately extends to the covert setup because the cardinality constraint is proved while preserving the P_X distribution.

We next specialize the covert Marton region to a family of distributions, where later we prove that this family exhausts the Marton region.

Pick auxiliary alphabets $\mathcal{U}$, $\mathcal{V}$, $\mathcal{T}$, and two symbols $u_1, u_2 \in \mathcal{U}$. We then let

$$\mathcal{B} \triangleq \mathcal{U}\setminus\{u_1,u_2\}$$

and $P_{UVT}^{\mathcal{B}}$ be an arbitrary distribution on $\mathcal{B}\times\mathcal{V}\times\mathcal{T}$, and

$$f_{\mathcal{B}}: \mathcal{B}\times\mathcal{V}\times\mathcal{T}\to\mathcal{X}$$

be an arbitrary deterministic mapping.

Further, we let P_X^1 and P_X^2 be two pmfs on $\mathcal{X}\setminus\{x_0\}$, and choose nonnegative numbers $\nu_0, \nu_1, \nu_2 \ge 0$ satisfying

$$\nu_0+\nu_1+\nu_2 = 1. \quad (14)$$

Fix arbitrary $\rho_1, \rho_2 > 0$ such that $\rho_1+\rho_2 = 1$, and let finally μ_n be a sequence that tends to 0 in the order of $\sqrt{1/n}$; the sequence is made precise in the proof of Theorem 2.

We now describe the joint distribution on (U,V,T,X) on which we evaluate the covert Marton region in (10)

$$P_U(u_1) = \rho_1(1-\nu_0\mu_n), \tag{15a}$$
$$P_U(u_2) = \rho_2(1-\nu_0\mu_n), \tag{15b}$$
$$P_U(u) = \nu_0\mu_n P_U^{\mathcal{B}}(u), \qquad u \in \mathcal{B}. \tag{15c}$$

When $U=u_1$, we take T to be constant, set $V=X$, and choose

$$P_{X|U}(x|u_1) = \begin{cases} \dfrac{\nu_1}{\rho_1}\dfrac{\mu_n}{1-\nu_0\mu_n}P_X^1(x), & x \neq x_0, \\ 1-\dfrac{\nu_1}{\rho_1}\dfrac{\mu_n}{1-\nu_0\mu_n}, & x = x_0. \end{cases} \tag{15d}$$

When $U=u_2$, we take V to be constant, set $T=X$, and choose

$$P_{X|U}(x|u_2) = \begin{cases} \dfrac{\nu_2}{\rho_2}\dfrac{\mu_n}{1-\nu_0\mu_n}P_X^2(x), & x \neq x_0, \\ 1-\dfrac{\nu_2}{\rho_2}\dfrac{\mu_n}{1-\nu_0\mu_n}, & x = x_0. \end{cases} \tag{15e}$$

For $u \in \mathcal{B}$, the conditional distribution of (V,T) is $P^{\mathcal{B}}_{VT|U=u}(\cdot,\cdot)$ and

$$X = f_{\mathcal{B}}(u,V,T). \tag{15f}$$

For the above parametrization, let $\tilde{\mathcal{L}}^{(\mathrm{I})}$ be the set of all nonnegative pairs (L_1,L_2) satisfying

$$L_1 \le \sqrt{\frac{2}{\chi_2(\boldsymbol{\nu})}}\Bigg[\nu_1 \sum_{x\neq x_0} P_X^1(x) D\left(P_{Y_1|X}(\cdot|x)\middle\|P_{Y_1|X}(\cdot|x_0)\right) + \nu_0\left[I^{\mathcal{B}}(U,V;Y_1) + D\left(P^{\mathcal{B}}_{Y_1}\middle\|P_{Y_1|X}(\cdot|x_0)\right)\right]\Bigg], \tag{16a}$$

$$L_2 \le \sqrt{\frac{2}{\chi_2(\boldsymbol{\nu})}}\Bigg[\nu_2 \sum_{x\neq x_0} P_X^2(x) D\left(P_{Y_2|X}(\cdot|x)\middle\|P_{Y_2|X}(\cdot|x_0)\right) + \nu_0\left[I^{\mathcal{B}}(U,T;Y_2) + D\left(P^{\mathcal{B}}_{Y_2}\middle\|P_{Y_2|X}(\cdot|x_0)\right)\right]\Bigg], \tag{16b}$$

and

$$\begin{aligned} L_1 + L_2 \le \sqrt{\frac{2}{\chi_2(\boldsymbol{\nu})}}\Bigg[&\nu_1 \sum_{x\neq x_0} P_X^1(x) D\left(P_{Y_1|X}(\cdot|x)\middle\|P_{Y_1|X}(\cdot|x_0)\right) \\ &+\nu_2 \sum_{x\neq x_0} P_X^2(x) D\left(P_{Y_2|X}(\cdot|x)\middle\|P_{Y_2|X}(\cdot|x_0)\right) \\ &+\nu_0 \min\Big\{I^{\mathcal{B}}(U;Y_1) + D\left(P^{\mathcal{B}}_{Y_1}\middle\|P_{Y_1|X}(\cdot|x_0)\right), \\ &\qquad I^{\mathcal{B}}(U;Y_2) + D\left(P^{\mathcal{B}}_{Y_2}\middle\|P_{Y_2|X}(\cdot|x_0)\right)\Big\} \\ &+\nu_0 I^{\mathcal{B}}(V;Y_1|U) + \nu_0 I^{\mathcal{B}}(T;Y_2|U) \\ &-\nu_0 I^{\mathcal{B}}(V;T|U)\Bigg]. \end{aligned} \tag{16c}$$

Here all mutual information quantities carrying the superscript $\mathcal{B}$ are calculated with respect to

$$P^{\mathcal{B}}_{UVT}(u,v,t)P_{Y_1Y_2Z|X}\big(y_1,y_2,z|f_{\mathcal{B}}(u,v,t)\big). \tag{17}$$

Moreover,

$$\chi_2(\boldsymbol{\nu}) \triangleq \chi_2\left(\nu_1 P_Z^1 + \nu_2 P_Z^2 + \nu_0 P_Z^{\mathcal{B}}\middle\|Q_0\right), \tag{18}$$

where

$$P_Z^1(z) \triangleq \sum_{x\neq x_0} P_X^1(x) P_{Z|X}(z|x), \tag{19}$$
$$P_Z^2(z) \triangleq \sum_{x\neq x_0} P_X^2(x) P_{Z|X}(z|x), \tag{20}$$
$$P_Z^{\mathcal{B}}(z) \triangleq \sum_{u\in\mathcal{B}}\sum_{u,v,t} P^{\mathcal{B}}_{UVT}(v,t|u) P_{Z|X}\big(z|f_{\mathcal{B}}(u,v,t)\big), \tag{21}$$

and $\chi_2(P\|Q)$ denotes the χ_2-distance between P and Q.

Theorem 2. *The region $\tilde{\mathcal{L}}^{(I)}$ is included in the covert Marton region $\mathcal{L}_\delta^{(\mathsf{M})}$.*

Using again the proof technique in [11] it can be shown that the region $\tilde{\mathcal{L}}^{(I)}$ is unchanged if the auxiliary alphabets are restricted to

$$|\mathcal{B}| \le |\mathcal{X}| + 4, \qquad |\mathcal{V}| \le |\mathcal{X}|, \qquad |\mathcal{T}| \le |\mathcal{X}|, \tag{22}$$

and therefore $|\mathcal{U}| \le |\mathcal{X}| + 6$.

Theorem 3. *The covert Marton region coincides with the region $\tilde{\mathcal{L}}^{(I)}$:*

$$\tilde{\mathcal{L}}^{(I)} = \mathcal{L}_\delta^{(\mathsf{M})}, \quad \forall \delta > 0. \tag{23}$$

III. Proof of Theorem 1

The achievability argument follows the direct part of the degraded-BC analysis [9], and is again based on the one-shot Marton bound of [10]. We use the following notation for the mutual information random variables [12], [13]:

$$i_{U;Y_k}(U;Y_k) = \log\frac{P_{Y_k|U}(Y_k|U)}{P_{Y_k}(Y_k)} \tag{24a}$$
$$i_{V;T|U}(V;T|U) = \log\frac{P_{V|U,T}(V|U,T)}{P_{V|U}(V|U)} \tag{24b}$$

and similarly for $i_{U;Y_2}(U;Y_2)$ etc.

For each n, take an n-dependent distribution $P^{(n)}_{UVT}$ and deterministic mapping $X = f_n(U,V,T)$ satisfying (10d), and use their n-fold memoryless extension $P^{(n)}_{U^nV^nT^nX^n}$. (In the following sometimes we omit the superscript (n) for ease of readability.) The one-shot Marton result in [10], the assumption that P_X satisfies (10d), and the union bound allow us to conclude that for any $\tilde{\gamma} > 0$ and $\ell_{1,p}, \ell_{2,p}, \ell_{1,c}, \ell_{2,c}, r_1', r_2' \ge 0$ with $\ell_{1,p} + \ell_{1,c} = \ell_1$ and $\ell_{2,p} + \ell_{2,c} = \ell_2$, there exists

an $(n, 2^{\ell_1}, 2^{\ell_2}, \epsilon, \delta)$ covert code for the BC if the following inequality is satisfied

$$\begin{aligned}\epsilon \leq\ & 2\exp(-\sqrt{n}\tilde{\gamma}) + \exp^{-\exp(\sqrt{n}\tilde{\gamma})} \\ &+P\left[\frac{1}{\sqrt{n}} i_{U^n,V^n;Y_1^n}(U^n,V^n;Y_1^n) \leq \frac{\ell_1+\ell_{2,c}+r_1'}{\sqrt{n}} + \tilde{\gamma}\right] \\ &+P\left[\frac{1}{\sqrt{n}} i_{U^n,T^n;Y_2^n}(U^n,T^n;Y_2^n) \leq \frac{\ell_{1,c}+\ell_2+r_2'}{\sqrt{n}} + \tilde{\gamma}\right] \\ &+P\left[\frac{1}{\sqrt{n}} i_{V^n;Y_1^n|U^n}(V^n;Y_1^n|U^n) \leq \frac{\ell_{1,p}+r_1'}{\sqrt{n}} + \tilde{\gamma}\right] \\ &+P\left[\frac{1}{\sqrt{n}} i_{T^n;Y_2^n|U^n}(T^n;Y_2^n|U^n) \leq \frac{\ell_{2,p}+r_2'}{\sqrt{n}} + \tilde{\gamma}\right] \\ &+P\left[\frac{1}{\sqrt{n}} i_{V^n;T^n|U^n}(V^n;T^n|U^n) \geq \frac{r_1'+r_2'}{\sqrt{n}} - 2\tilde{\gamma}\right] \end{aligned} \tag{25}$$

Following the steps in [9], it can be shown that the random quantities $\frac{1}{\sqrt{n}} i_{U^n;Y_1^n}(U^n;Y_1^n)$ etc. converge in probability to the corresponding single-letter information functions.[2] Precisely,

$$\frac{1}{\sqrt{n}} i_{U^n;Y_k^n}(U^n;Y_k^n) \longrightarrow \sqrt{n} I(U;Y_k) \quad \text{in prob.}, \quad k \in \{1,2\}, \tag{26a}$$

$$\frac{1}{\sqrt{n}} i_{U^n,V^n;Y_1^n}(U^n,V^n;Y_1^n) \longrightarrow \sqrt{n} I(U,V;Y_1) \quad \text{in prob.} \tag{26b}$$

$$\frac{1}{\sqrt{n}} i_{U^n,T^n;Y_2^n}(U^n,T^n;Y_2^n) \longrightarrow \sqrt{n} I(U,T;Y_2) \quad \text{in prob.} \tag{26c}$$

From the above limits we directly obtain the following related convergence results:

$$\begin{aligned}\frac{1}{\sqrt{n}} i_{V^n;Y_1^n|U^n}(V^n;Y_1^n|U^n) &\longrightarrow \sqrt{n} I(V;Y_1|U) \quad \text{in prob.} \\ \frac{1}{\sqrt{n}} i_{T^n;Y_2^n|U^n}(T^n;Y_2^n|U^n) &\longrightarrow \sqrt{n} I(T;Y_2|U) \quad \text{in prob.}\end{aligned} \tag{26d}$$

We are left with proving the convergence of the information density $\frac{1}{\sqrt{n}} i_{V^n;T^n|U^n}$ (Marton binning term). We first note that for each n

$$\mathbb{E}\left[\frac{1}{\sqrt{n}} i_{V^n;T^n|U^n}(V^n;T^n|U^n)\right] = \sqrt{n} I(V;T|U). \tag{27}$$

We will next prove that as $n \to \infty$

$$\mathrm{var}\left(\frac{1}{\sqrt{n}} i_{V^n;T^n|U^n}(V^n;T^n|U^n)\right) \longrightarrow 0, \tag{28}$$

which by Chebyshev's inequality then establishes the desired convergence in probability:

$$\frac{1}{\sqrt{n}} i_{V^n;T^n|U^n}(V^n;T^n|U^n) \longrightarrow \sqrt{n} I(V;T|U) \quad \text{in prob.} \tag{29}$$

[2]The analysis of the information quantities $\frac{1}{\sqrt{n}} i_{U^n;Y_k^n}(U^n;Y_k^n)$ is exactly as in [9]; for the analysis of $\frac{1}{\sqrt{n}} i_{U^n,V^n;Y_1^n}(U^n,V^n;Y_1^n)$ and $\frac{1}{\sqrt{n}} i_{U^n,T^n;Y_2^n}(U^n,T^n;Y_2^n)$ the same proof steps apply if one considers the pair (U,T) as a single auxiliary.

Indeed we have for the variance

$$\begin{aligned}&\mathsf{var}\left(\frac{1}{\sqrt{n}} i_{V^n;T^n|U^n}(V^n;T^n|U^n)\right) \\ &= \frac{1}{n}\sum_{i=1}^{n} \mathsf{var}\left(\log\frac{P_{V|U,T}(V_i|U_i,T_i)}{P_{V|U}(V_i|U_i)}\right) &(30)\\ &= \mathsf{var}\left(\log\frac{P_{V|U,T}(V|U,T)}{P_{V|U}(V|U)}\right) &(31)\\ &= \mathsf{var}\left(i_{V;T|U}(V;T|U)\right) &(32)\\ &\leq \mathsf{E}\left[\left(i_{V;T|U}(V;T|U)\right)^2\right]. &(33)\end{aligned}$$

Notice next that we can restrict attention to pmfs $P_{UVT}^{(n)}$ so that $I(V;T|U) \to 0$ as $n \to \infty$ because otherwise the Marton covering term makes the achievable region void as it dominates the remaining terms. However, when $I(V;T|U) \to 0$, then for all tuples (u,v,t) where the limit of $P_{UVT}^{(n)}(u,v,t)$ is strictly positive, the ratio $P_{VT|U}^{(n)}(v,t|u)/(P_{V|U}^{(n)}(v|u)P_{T|U}^{(n)}(t|u)) \to 1$ and both $i_{V;T|U}$ and $i_{V;T|U}^2(v;t|u)$ vanish in the limit as $n \to \infty$. The asymptotic variance is thus determined by the triples (u,v,t) with vanishing $P_{UVT}^{(n)}(u,v,t)$. For these triples we have

$$\begin{aligned}&P_{UVT}(u,v,t) i^2(V;T|U)(v;t|u) \\ &\leq P_{UVT}(u,v,t)\log^2\frac{1}{P_{UVT}(u,v,t)} \to 0,\end{aligned} \tag{34}$$

where the limiting behaviour holds because $t\log^2(1/t) \to 0$ as $t \to 0$. By the finiteness of the alphabets, we conclude that in the limit $n \to \infty$

$$\mathsf{var}\big(i_{V;T|U}(V;T|U)\big) \longrightarrow 0. \tag{35}$$

We have shown that for any sequence of per-symbol distributions $P_{UVTX}^{(n)}$ with marginals $P_X^{(n)}$ satisfying (10d), the random variables in (25) converge in probability to the corresponding mutual information functions. This implies that for any $\tilde{\gamma} > 0$ and any tuple $\ell_{1,c}, \ell_{2,c}, \ell_{1,p}, \ell_{2,p}, r_1', r_2' \geq 0$ with $\ell_1 = \ell_{1,c} + \ell_{1,p}$ and $\ell_2 = \ell_{2,c} + \ell_{2,p}$ satisfying the following constraints

$$\frac{\ell_1+\ell_{2,c}+r_1'}{\sqrt{n\delta}} < \sqrt{n/\delta} I(U,V;Y_1) - \tilde{\gamma}/\delta \tag{36a}$$

$$\frac{\ell_{1,c}+\ell_2+r_2'}{\sqrt{n\delta}} < \sqrt{n/\delta} I(U,T;Y_2) - \tilde{\gamma}/\delta \tag{36b}$$

$$\frac{\ell_{1,p}+r_1'}{\sqrt{n\delta}} < \sqrt{n/\delta} I(V;Y_1|U) - \tilde{\gamma}/\delta \tag{36c}$$

$$\frac{\ell_{2,p}+r_2'}{\sqrt{n\delta}} < \sqrt{n/\delta} I(T;Y_2|U) - \tilde{\gamma}/\delta \tag{36d}$$

$$\frac{r_1'+r_2'}{\sqrt{n\delta}} > \sqrt{n/\delta} I(V;T|U) + 2\tilde{\gamma}/\delta, \tag{36e}$$

there exists an $(n, \nu_1 = 2^{\ell_1}, \nu_2 = 2^{\ell_2}, \epsilon, \delta)$ covert code for the BC.

For $k \in \{1,2\}$ replace $\ell_{k,p}$ by $\ell_k - \ell_{k,c}$ and set the covert rate $L_k := \frac{\ell_1}{\sqrt{n\delta}}$. Let then $\tilde{\gamma} \to 0$ and apply a Fourier-Motzkin elimination step to get rid of the auxiliary

parameters $\ell_{1,c}, \ell_{2,c}, r_1', r_2'$. This establishes the desired result in Theorem 1.

IV. Proof of Theorem 2

For the parametrization in (15), all the mutual informations occurring in $\mathcal{L}_\delta^{(M)}$ vanish as $\mu_n \to 0$. We are thus interested in the rate of convergence, which we obtain using a Taylor expansion of all the mutual informations occurring in $\mathcal{L}_\delta^{(M)}$ at $\mu_n = 0$.

According to the calculations in Appendix A, we obtain the following Taylor expansions

$$\begin{aligned} I(U;Y_1) &= \mu_n\nu_0 \left[I^{\mathcal{B}}(U;Y_1) + D\left(P_{Y_1}^{\mathcal{B}} \middle\| P_{Y_1|X}(\cdot|x_0)\right)\right] \\ &\quad + o(\mu_n), \qquad (37) \\ I(V;Y_1|U) &= \mu_n\nu_1 \sum_{x\neq x_0} P_X^1(x) D\left(P_{Y_1|X}(\cdot|x) \middle\| P_{Y_1|X}(\cdot|x_0)\right) \\ &\quad + \mu_n\nu_0 I^{\mathcal{B}}(V;Y_1|U) + o(\mu_n), \qquad (38) \\ I(V;T|U) &= \mu_n\nu_0 I^{\mathcal{B}}(V;T|U) + o(\mu_n). \qquad (39) \end{aligned}$$

By symmetric arguments, we can also obtain:

$$\begin{aligned} I(U;Y_2) &= \mu_n\nu_0 \left[I^{\mathcal{B}}(U;Y_2) + D\left(P_{Y_2}^{\mathcal{B}} \middle\| P_{Y_2|X}(\cdot|x_0)\right)\right] \\ &\quad + o(\mu_n) \qquad (40) \end{aligned}$$

and

$$\begin{aligned} I(T;Y_2|U) &= \mu_n\nu_2 \sum_{x\neq x_0} P_X^2(x) D\left(P_{Y_2|X}(\cdot|x) \middle\| P_{Y_2|X}(\cdot|x_0)\right) \\ &\quad + \mu_n\nu_0 I^{\mathcal{B}}(V;Y_2|U) + o(\mu_n). \qquad (41) \end{aligned}$$

It remains to evaluate the covertness constraint. By construction, the warden output distribution satisfies

$$P_Z = (1-\mu_n)Q_0 + \mu_n\left(\nu_1 P_Z^1 + \nu_2 P_Z^2 + \nu_0 P_Z^{\mathcal{B}}\right). \qquad (42)$$

Therefore,

$$\begin{aligned} D(P_Z\|Q_0) &= \frac{\mu_n^2}{2}\chi_2\left(\nu_1 P_Z^1 + \nu_2 P_Z^2 + \nu_0 P_Z^{\mathcal{B}} \middle\| Q_0\right) + o(\mu_n^2) \\ &= \frac{\mu_n^2}{2}\chi_2(\boldsymbol{\nu}) + o(\mu_n^2). \qquad (43) \end{aligned}$$

Choosing

$$\mu_n = \sqrt{\frac{2\delta}{n\cdot\chi_2(\boldsymbol{\nu})}} \qquad (44)$$

saturates the covertness constraint asymptotically. Since

$$\lim_{n\to\infty}\sqrt{\frac{n}{\delta}}\,\mu_n = \sqrt{\frac{2}{\chi_2(\boldsymbol{\nu})}}, \qquad (45)$$

combining all the above considerations and letting $n \to \infty$ (and thus $\mu_n \to 0$) establishes the theorem.

V. Proof of Theorem 3

We construct here a sufficiently general joint distribution $P_{U,V,TX}$ that adheres to (10d). As a close inspection of our proof reveals, more general distributions do not change the region since the constraints on L_1 and L_2 only depend on the limiting points of these more general distributions, which we exhaust in our framework.

Let $\mathcal{B}$ be a proper subset of the auxiliary alphabet $\mathcal{U}$ and $\mathcal{A}$ its complement. Let $P_U^{\mathcal{B}}$ and $P_U^{\mathcal{A}}$ be general distributions on $\mathcal{B}$ and $\mathcal{A}$, respectively, and set

$$\hat{P}_U(u) = (1-\mu_n)P_U^{\mathcal{A}}(u) + \mu_n P_U^{\mathcal{B}}(u), \qquad (46)$$

where μ_n is some sequence that tends to 0 as $n \to \infty$ at a rate which will be determined later.

Now, fix a general mapping $f\colon \mathcal{U}\times\mathcal{V}\times\mathcal{T} \to \mathcal{X}$, and for each $u \in \mathcal{B}$ let $P_{V,T|U}^{\mathcal{B}}(\cdot|u)$ be a conditional distribution on $\mathcal{V}\times\mathcal{T}$. Assume also a general distribution $P_{VT|U}^{\mathcal{A}}(\cdot|u)$ for each $u \in \mathcal{A}$, and for each $u \in \mathcal{A}$, partition the satellite alphabets as

$$\mathcal{V} = \mathcal{V}_0(u) \cup \mathcal{V}_1(u), \qquad \mathcal{T} = \mathcal{T}_0(u) \cup \mathcal{T}_1(u), \qquad (47)$$

where the symbols in the subscript-0 sets have nonvanishing conditional probabilities and those in the subscript-1 sets have vanishing probabilities. As $n \to \infty$, the joint pmf P_{UVT} over $\mathcal{U}\times\mathcal{V}\times\mathcal{T}$ concentrates on the set of tuples

$$\mathcal{S}_0 \triangleq \{(u,v,t)\colon u \in \mathcal{A},\ v \in \mathcal{V}_0(u),\ t \in \mathcal{T}_0(u)\}. \qquad (48)$$

As a consequence, the covertness constraint (10d), implies that the function f maps every triple $(u,v,t) \in \mathcal{S}_0$ to the innocent symbol x_0. Moreover, by continuity of mutual information, the conditional mutual information $I(V;T|U)$ converges to the conditional mutual information of the pmf P_{UVT} restricted to $\mathcal{S}_0$. Since under the covertness constraint all other mutual information terms in Marton's region also tend to 0 as $n \to \infty$, we can conclude that the constraints on L_1 and L_2 are degenerate unless the limiting distribution of $P_{VT|U}$ factorizes over $\mathcal{S}_0$:

$$\begin{aligned} \lim_{n\to\infty} P_{VT|U=u}(v,t) &= P_{V|U}^{00}(v|u)P_{T|U}^{00}(t|u), \\ &\qquad \forall u \in \mathcal{A}, v \in \mathcal{V}_0(u), t \in \mathcal{T}_0(u), \qquad (49) \end{aligned}$$

for some $P_{V|U}^{00}(\cdot|u)$ and $P_{T|U}^{00}(\cdot|u)$ distributions for all $u \in \mathcal{A}$.

We thus restrict to distributions of the form[3]

$$P_{VT|U}^{\mathcal{A}}(v,t|u) = \begin{cases} (1-\mu_n e_u)P_{VT|U=u}^{00}(v,t), & (v,t) \in \mathcal{V}_0(u)\times\mathcal{T}_0(u), \\ \mu_n\alpha_u P_{VT|U=u}^{10}(v,t), & (v,t) \in \mathcal{V}_1(u)\times\mathcal{T}_0(u), \\ \mu_n\beta_u P_{VT|U=u}^{01}(v,t), & (v,t) \in \mathcal{V}_0(u)\times\mathcal{T}_1(u), \\ \mu_n\gamma_u P_{VT|U=u}^{11}(v,t), & (v,t) \in \mathcal{V}_1(u)\times\mathcal{T}_1(u), \end{cases} \qquad (50)$$

for some $\alpha_u, \beta_u, \gamma_u \geq 0$ and $e_u := \alpha_u + \beta_u + \gamma_u$ and conditional pmfs $P_{VT|U=u}^{01}$, $P_{VT|U=u}^{10}$, $P_{VT|U=u}^{11}$, and $P_{VT|U=u}^{00} = P_{V|U=u}^{00}P_{T|U=u}^{00}$.

[3]Here, we could consider more general distributions. A close inspection of the forthcoming proof steps however shows that the final result will only depend on the accumulation points of the pmfs P^{00}, P^{01}, and P^{10} and factors α_u and β_u.

Under this parametrization, the mutual information terms in the covert Marton region involving the outputs decay in the order of μ_n. (This is easily checked following the calculations in our Appendix A or in [9].) Hence, for the covert Marton region to be non-degenerate we need all contributions $I(V;T|U=u)$, for $u \in \mathcal{A}$, to be at most of order μ_n. However, we observe that for pairs $(v,t) \in \mathcal{V}_1(u) \times \mathcal{T}_1(u)$ the contribution is

$$\begin{aligned} &P_{VT|U=u}(v,t) \log \frac{P_{VT|U=u}(v,t)}{P_{V|U=u}(v) P_{T|U=u}(t)} \\ &= \Theta\left(\mu_n \log \frac{\mu_n}{\mu_n^2}\right) = \Theta\left(\mu_n \log \frac{1}{\mu_n}\right) \end{aligned} \tag{51}$$

which decreases to zero more slowly than μ_n and thus leads to a completely degenerate covert Marton region.

For $u \in \mathcal{A}$, we can thus restrict without loss in optimality to distributions with zero probability over the pairs $\mathcal{V}_1(u) \times \mathcal{T}_1(u)$, i.e.,

$$\gamma_u = 0 \tag{52}$$

in the above parametrization (50). A consequence of this is that for $u \in \mathcal{A}$:

$$\begin{aligned} &f(u,v,t) \neq x_0 \\ &\quad \Rightarrow \quad (v,t) \in (\mathcal{V}_0(u) \times \mathcal{T}_1(u)) \cup (\mathcal{V}_1(u) \times \mathcal{T}_0(u)). \end{aligned} \tag{53}$$

We next derive the first-order behavior of the mutual information expressions in the covert Marton region for the parametrization given by (46), (50), and (52), and a fully general $P^{\mathcal{B}}_{VT|U}$. Start by noting that for any $u \in \mathcal{A}$ and $x \neq x_0$ by (53):

$$P_{X|U}(x|u) = \sum_{(v,t):\ f(u,v,t)=x} P_{VT|U=u}(v,t) \tag{54}$$

$$= \Theta(\mu_n). \tag{55}$$

As a consequence, for all $u \in \mathcal{A}$,

$$\lim_{n\to\infty} P_{X|U}(x|u) = \mathbf{1}\{x = x_0\} \tag{56}$$

$$\lim_{n\to\infty} P_{Y_k|U}(y_k|u) = P_{Y_k|X}(y_k|x_0), \quad k \in \{1,2\}. \tag{57}$$

Since also $P_{Y_k}(y_k)$ tends to $P_{Y_k|X}(y_k|x_0)$, for $k \in \{1,2\}$, we have:

$$\lim_{n\to\infty} \log \frac{P_{Y_k|U}(y_k|u)}{P_{Y_k}(y_k)} = \lim_{n\to\infty} \log \frac{P_{Y_k|U}(y_k|u)}{P_{Y_k|X}(y_k|x_0)} = 0. \tag{58}$$

As shown in Appendix A-B, due to (58) the symbols $u \in \mathcal{A}$ do not contribute to $I(U;Y_k)$ but only symbols $u \in \mathcal{B}$ do and

$$\begin{aligned} I(U;Y_k) = \mu_n \left[I^{\mathcal{B}}(U;Y_k) + D\left(P^{\mathcal{B}}_{Y_k} \middle\| P_{Y_k|X}(\cdot|x_0)\right)\right] \\ + o(\mu_n). \end{aligned} \tag{59}$$

To analyze the mutual information expression $I(U,T;Y_2)$, note that for $u \in \mathcal{A}$ and $t \in \mathcal{T}_0(u)$, for any $x \in \mathcal{X}\backslash\{x_0\}$:

$$P_{X|T,U}(x|t,u) = \sum_{v:\ f(u,v,t)=x} P_{V|T,U}(v|t,u) \tag{60}$$

$$= \sum_{v \in \mathcal{V}_1:\ f(u,v,t)=x} \frac{P_{VT|U=u}(v,t)}{P_{T|U}(t|u)} \tag{61}$$

$$= O(\mu_n), \tag{62}$$

where the second equality holds by (53) and the last equality because for $t \in \mathcal{T}_0(u)$ and $v \in \mathcal{V}_1(u)$ we have $P_{VT|U=u}(v,t) = \Theta(\mu_n)$ while $P_{T|U=u}(t) = \Theta(1)$. As a consequence, for $t \in \mathcal{T}_0(u)$:

$$\lim_{n\to\infty} P_{X|T,U}(x|t,u) = \mathbf{1}\{x = x_0\} \tag{63}$$

$$\lim_{n\to\infty} P_{Y_2|T,U}(y_2|t,u) = P_{Y_2|X}(y_2|x_0), \tag{64}$$

which implies

$$\lim_{n\to\infty} \log \frac{P_{Y_2|T,U}(y_2|t,u)}{P_{Y_2}(y_2)} = \lim_{n\to\infty} \log \frac{P_{Y_2|T,U}(y_2|t,u)}{P_{Y_2|X}(y_2|x_0)} = 0. \tag{65}$$

By contrast, for $t \in \mathcal{T}_1(u)$ we have for any $x \in \mathcal{X}\backslash\{x_0\}$:

$$P_{X|T,U}(x|t,u)$$

$$= \sum_{\substack{v \in \mathcal{V}_0(u):\\ f(u,v,t)=x}} \frac{P_{VT|U=u}(v,t)}{P_{T|U}(t|u)} \tag{66}$$

$$= \sum_{\substack{v \in \mathcal{V}_0(u):\\ f(u,v,t)=x}} \frac{P^{01}_{VT|U=u}(v,t)}{\sum_{v' \in \mathcal{V}_0(u)} P^{01}_{VT|U=u}(v',t)} + o(1), \tag{67}$$

and as a consequence, defining

$$\begin{aligned} &P^{01}_{Y_2|T,U=u}(y_2|t) \\ &\quad \triangleq \sum_{v \in \mathcal{V}_0(u)} P_{Y_2|X}(y_2|f(u,v,t)) P^{01}_{V|T,U=u}(v|t) \end{aligned} \tag{68}$$

and

$$P^{01}_{V|T,U=u}(v|t) \triangleq \frac{P^{01}_{VT|U=u}(v,t)}{\sum_{v' \in \mathcal{V}_0(u)} P^{01}_{VT|U=u}(v',t)}, \tag{69}$$

we have for $t \in \mathcal{T}_1(u)$:

$$\lim_{n\to\infty} \log \frac{P_{Y_2|T,U}(y_2|t,u)}{P_{Y_2}(y_2)} = \lim_{n\to\infty} \log \frac{P^{01}_{Y_2|T,U=u}(y_2|t)}{P_{Y_2|X}(y_2|x_0)}. \tag{70}$$

As shown by the derivative calculations in Appendix A-B, the fact that t-values in $\mathcal{T}_0(u)$ have limit zero in (65), while t-values in $\mathcal{T}_1(u)$ have a strictly nonzero limit in (70), implies that only the t-values in $\mathcal{T}_1(u)$ contribute to the Taylor expansion of $I(U,T;Y_2)$ but not the t-values in $\mathcal{T}_0$. More precisely, based on the derivatives obtained in Appendix A-B, the following Taylor expansion holds:

$$\begin{aligned} &I(U,T;Y_2) \\ &= \mu_n \sum_{\substack{u \in \mathcal{A},\\ t \in \mathcal{T}_1(u)}} \beta_u P^{\mathcal{A}}_U(u) P^{01}_{T|U=u}(t) \\ &\qquad \cdot D\left(P^{01}_{Y_2|T,U=u}(\cdot|t) \| P_{Y_2|X}(\cdot|x_0)\right) \\ &\quad + \nu_0 \left[I^{\mathcal{B}}(U,T;Y_2) + D\left(P^{\mathcal{B}}_{Y_2} \middle\| P_{Y_2|X}(\cdot|x_0)\right)\right] + o(\mu_n) \end{aligned} \tag{71}$$

$$\begin{aligned} &\leq \mu_n \sum_{u \in \mathcal{A}, x \in \mathcal{X}} \beta_u P^{\mathcal{A}}_U(u) P^{01}_{X|U=u}(x) \\ &\qquad \cdot D\left(P^{01}_{Y_2|X}(\cdot|x) \| P_{Y_2|X}(\cdot|x_0)\right) \\ &\quad + \nu_0 \left[I^{\mathcal{B}}(U,T;Y_2) + D\left(P^{\mathcal{B}}_{Y_2} \middle\| P_{Y_2|X}(\cdot|x_0)\right)\right] + o(\mu_n), \end{aligned} \tag{72}$$

where the inequality holds by the convexity of KL divergence and we defined

$$P^{01}_{X|U=u}(x) \triangleq \sum_{\substack{v\in\mathcal{V}_0(u),t\in\mathcal{T}_1(u):\\ f(u,v,t)=x}} P^{01}_{VT|U=u}(v,t). \tag{73}$$

In the same way, we can likewise conclude that

$$\begin{aligned} &I(U,V;Y_1)\\ &\le \mu_n \sum_{\substack{u\in\mathcal{A},\\ x\in\mathcal{X}}} \alpha_u P^{\mathcal{A}}_U(u) P^{10}_{X|U=u}(x) D\left(P^{10}_{Y_1|X}(\cdot|x)\|P_{Y_1|X}(\cdot|x_0)\right)\\ &\quad+\nu_0\left[I^{\mathcal{B}}(U,V;Y_1)+D\left(P^{\mathcal{B}}_{Y_1}\middle\|P_{Y_1|X}(\cdot|x_0)\right)\right]\\ &\quad+o(\mu_n), \end{aligned} \tag{74}$$

when $P^{10}_{X|U=u}(x)$ is defined appropriately.

Notice next that by simply omitting the $\mathcal{A}$-contributions, we obtain

$$I(V;T|U) \ge \mu_n I^{\mathcal{B}}(V;T|U). \tag{75}$$

Combine now bounds (72), (74), and (75), and follow the same steps as in (42)–(45) to account for the covertness constraint. Identifying in the resulting region

$$\nu_1 P^1_X(x) = \sum_{u\in\mathcal{A}} \alpha_u P^{\mathcal{A}}_U(u) P^{10}_{X|U=u}(x) \tag{76}$$

$$\nu_2 P^2_X(x) = \sum_{u\in\mathcal{A}} \beta_u P^{\mathcal{A}}_U(u) P^{01}_{X|U=u}(x), \tag{77}$$

we can conclude that the general covert Marton region $\mathcal{L}^{(M)}$ cannot be larger than the region $\tilde{\mathcal{L}}^{(I)}$. This concludes the desired proof.

Appendix A
Derivative Calculations

We derive the first-order expansions of the mutual information terms appearing in the covert Marton region.

A. Parametrization in (15)

Start by analyzing the derivative

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}\mu} I(U;Y_1) &= \sum_{u,y_1} \frac{\mathrm{d}P_{UY_1}(u,y_1)}{\mathrm{d}\mu} \log \frac{P_{UY_1}(u,y_1)}{P_U(u)P_{Y_1}(y_1)}\\ &\quad+\sum_{u,y_1} P_{UY_1}(u,y_1) \frac{\mathrm{d}}{\mathrm{d}\mu} \log \frac{P_{UY_1}(u,y_1)}{P_U(u)P_{Y_1}(y_1)}. \end{aligned} \tag{78}$$

The second term in (78) can be expanded as

$$\begin{aligned} &\sum_{u,y_1} P_{UY_1}(u,y_1) \frac{\mathrm{d}}{\mathrm{d}\mu} \log \frac{P_{UY_1}(u,y_1)}{P_U(u)P_{Y_1}(y_1)}\\ &\quad= \sum_{u,y_1} P_{UY_1}(u,y_1) \frac{1}{P_{UY_1}(u,y_1)} \frac{\mathrm{d}P_{UY_1}(u,y_1)}{\mathrm{d}\mu}\\ &\qquad- \sum_{u,y_1} P_{UY_1}(u,y_1) \frac{1}{P_U(u)} \frac{\mathrm{d}P_U(u)}{\mathrm{d}\mu}\\ &\qquad- \sum_{u,y_1} P_{UY_1}(u,y_1) \frac{1}{P_{Y_1}(y_1)} \frac{\mathrm{d}P_{Y_1}(y_1)}{\mathrm{d}\mu}. \end{aligned} \tag{79}$$

The first term on the right-hand side of (79) is

$$\begin{aligned} \sum_{u,y_1} \frac{\mathrm{d}P_{UY_1}(u,y_1)}{\mathrm{d}\mu} &= \frac{\mathrm{d}}{\mathrm{d}\mu} \sum_{u,y_1} P_{UY_1}(u,y_1)\\ &= \frac{\mathrm{d}}{\mathrm{d}\mu} 1 = 0. \end{aligned} \tag{80}$$

For the second term, using $\sum_{y_1} P_{UY_1}(u,y_1) = P_U(u)$,

$$\begin{aligned} &\sum_{u,y_1} P_{UY_1}(u,y_1) \frac{1}{P_U(u)} \frac{\mathrm{d}P_U(u)}{\mathrm{d}\mu}\\ &\quad= \sum_u \frac{\mathrm{d}P_U(u)}{\mathrm{d}\mu}\\ &\quad= \frac{\mathrm{d}}{\mathrm{d}\mu} \sum_u P_U(u) = 0. \end{aligned} \tag{81}$$

Similarly, since $\sum_u P_{UY_1}(u,y_1) = P_{Y_1}(y_1)$,

$$\begin{aligned} &\sum_{u,y_1} P_{UY_1}(u,y_1) \frac{1}{P_{Y_1}(y_1)} \frac{\mathrm{d}P_{Y_1}(y_1)}{\mathrm{d}\mu}\\ &\quad= \sum_{y_1} \frac{\mathrm{d}P_{Y_1}(y_1)}{\mathrm{d}\mu}\\ &\quad= \frac{\mathrm{d}}{\mathrm{d}\mu} \sum_{y_1} P_{Y_1}(y_1) = 0. \end{aligned} \tag{82}$$

It follows from (79)–(82) that

$$\sum_{u,y_1} P_{UY_1}(u,y_1) \frac{\mathrm{d}}{\mathrm{d}\mu} \log \frac{P_{UY_1}(u,y_1)}{P_U(u)P_{Y_1}(y_1)} = 0, \tag{83}$$

and thus the second summand in (78) vanishes.

We turn attention to the first summand of (78) and explicitly compute $\frac{\mathrm{d}}{\mathrm{d}\mu} P_{UY_1}(u,y_1)$ for the three subsets of $\mathcal{U}$. For $u = u_1$,

$$\begin{aligned} P_{UY_1}(u_1,y_1) &= \left[\rho_1(1-\nu_0\mu)-\nu_1\mu\right] P_{Y_1|X}(y_1|x_0)\\ &\quad+\nu_1\mu \sum_{x\neq x_0} P^1_X(x) P_{Y_1|X}(y_1|x), \end{aligned} \tag{84}$$

and hence

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}\mu} P_{UY_1}(u_1,y_1) &= -[\rho_1\nu_0+\nu_1] P_{Y_1|X}(y_1|x_0)\\ &\quad+\nu_1 \sum_{x\neq x_0} P^1_X(x) P_{Y_1|X}(y_1|x). \end{aligned} \tag{85}$$

Similarly, for $u = u_2$,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}\mu} P_{UY_1}(u_2,y_1) &= -[\rho_2\nu_0+\nu_2] P_{Y_1|X}(y_1|x_0)\\ &\quad+\nu_2 \sum_{x\neq x_0} P^2_X(x) P_{Y_1|X}(y_1|x). \end{aligned} \tag{86}$$

Finally, for $u \in \mathcal{B}$,

$$\frac{\mathrm{d}}{\mathrm{d}\mu} P_{UY_1}(u,y_1) = \nu_0 P^{\mathcal{B}}_U(u) P^{\mathcal{B}}_{Y_1|U}(y_1|u), \qquad u\in\mathcal{B}. \tag{87}$$

We next evaluate the logarithmic term

$$\log \frac{P_{UY_1}(u,y_1)}{P_U(u)P_{Y_1}(y_1)} = \log \frac{P_{Y_1|U}(y_1|u)}{P_{Y_1}(y_1)} \tag{88}$$

at $\mu = 0$ for the three components of U.

Obviously, at $\mu = 0$ the output marginal is

$$P_{Y_1}(y_1)|_{\mu=0} = P_{Y_1|X}(y_1|x_0). \tag{89}$$

The same holds also conditioned on $U = u_1$,

$$P_{Y_1|U}(y_1|u_1)\big|_{\mu=0} = P_{Y_1|X}(y_1|x_0). \tag{90}$$

from which we can conclude that

$$\log \frac{P_{Y_1|U}(y_1|u_1)}{P_{Y_1}(y_1)}\bigg|_{\mu=0} = 0. \tag{91}$$

Similarly, for $u = u_2$

$$\log \frac{P_{Y_1|U}(y_1|u_2)}{P_{Y_1}(y_1)}\bigg|_{\mu=0} = 0. \tag{92}$$

Finally, for $u \in \mathcal{B}$, the conditional output distribution is independent of μ, and hence

$$\log \frac{P_{Y_1|U}(y_1|u)}{P_{Y_1}(y_1)}\bigg|_{\mu=0} = \log \frac{P^{\mathcal{B}}_{Y_1|U}(y_1|u)}{P_{Y_1|X}(y_1|x_0)}, \qquad u \in \mathcal{B}. \tag{93}$$

Combining all these calculations, we obtain for the first summand of (78):

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\mu} \sum_{u,y_1} P^{(\mu)}_{UY_1}(u,y_1) \log \frac{P_{Y_1|U}(y_1|u)}{P_{Y_1}(y_1)}\bigg|_{\mu=0} \\ = \nu_0 \sum_{u\in\mathcal{B}, y_1\in\mathcal{Y}_1} P^{\mathcal{B}}_U(u) P^{\mathcal{B}}_{Y_1|U}(y_1|u) \log \frac{P^{\mathcal{B}}_{Y_1|U}(y_1|u)}{P_{Y_1|X}(y_1|x_0)},\end{aligned} \tag{94}$$

and for the desired derivative

$$\begin{aligned}&\frac{\mathrm{d}}{\mathrm{d}\mu} I(U;Y_1)\bigg|_{\mu=0} \\ &= \nu_0 \sum_{u\in\mathcal{B}, y_1\in\mathcal{Y}_1} P^{\mathcal{B}}_U(u) P^{\mathcal{B}}_{Y_1|U}(y_1|u) \log \frac{P^{\mathcal{B}}_{Y_1|U}(y_1|u)}{P_{Y_1|X}(y_1|x_0)} &(95)\\ &= \nu_0 \sum_{u\in\mathcal{B}} P^{\mathcal{B}}_U(u) D(P^{\mathcal{B}}_{Y_1|U}(\cdot|u) \| P_{Y_1|X}(\cdot|x_0)) &(96)\\ &= \nu_0 \left[I^{\mathcal{B}}(U;Y_1) + D\left(P^{\mathcal{B}}_{Y_2} \middle\| P_{Y_2|X}(\cdot|x_0)\right)\right]. &(97)\end{aligned}$$

We next calculate the derivative of $I(V;Y_1|U)$. To this end, write

$$\begin{aligned}&\frac{\mathrm{d}}{\mathrm{d}\mu} I(V;Y_1|U)\bigg|_{\mu=0} \\ &= \rho_1 \frac{\mathrm{d}}{\mathrm{d}\mu}\left[(1-\nu_0\mu) I(V;Y_1|U=u_1)\right]\bigg|_{\mu=0} \\ &\quad + \rho_2 \frac{\mathrm{d}}{\mathrm{d}\mu}\left[(1-\nu_0\mu) I(V;Y_1|U=u_2)\right]\bigg|_{\mu=0} \\ &\quad + \nu_0 I^{\mathcal{B}}(V;Y_1|U),\end{aligned} \tag{98}$$

and remark that

$$I(V;Y_1|U=u_1) = I(X;Y_1|U=u_1) \tag{99}$$

and

$$I(V;Y_1|U=u_2) = 0. \tag{100}$$

By the standard calculations for covert communications, we then obtain:

$$\begin{aligned}&\rho_1 \frac{\mathrm{d}}{\mathrm{d}\mu}\left[(1-\nu_0\mu) I(X;Y_1|U=u_1)\right]\bigg|_{\mu=0} \\ &= \frac{\mathrm{d}}{\mathrm{d}\mu} \sum_{x\neq x_0, y_1} \nu_1 \mu P^1_X(x) P_{Y_1|X}(y_1|x) \log \frac{P_{Y_1|X}(y_1|x)}{P_{Y_1}(y_1)}\bigg|_{\mu=0} \\ &\quad + \frac{\mathrm{d}}{\mathrm{d}\mu}(\rho_1 - (\rho_1\nu_0 + \nu_1)\mu) P_{Y_1|X}(y_1|x_0) \\ &\qquad\qquad \cdot \log \frac{P_{Y_1|X}(y_1|x_0)}{P_{Y_1}(y_1)}\bigg|_{\mu=0} &(101)\\ &= \nu_1 \sum_{x\neq x_0} P^1_X(x) D\left(P_{Y_1|X}(\cdot|x) \middle\| P_{Y_1|X}(\cdot|x_0)\right), &(102)\end{aligned}$$

and can conclude that

$$\begin{aligned}&\frac{\mathrm{d}}{\mathrm{d}\mu} I(V;Y_1|U)\bigg|_{\mu=0} \\ &\quad = \nu_1 \sum_{x\neq x_0} P^1_X(x) D\left(P_{Y_1|X}(\cdot|x) \middle\| P_{Y_1|X}(\cdot|x_0)\right) \\ &\qquad + \nu_0 I^{\mathcal{B}}(V;Y_1|U).\end{aligned} \tag{103}$$

We conclude this section with the computation of the derivative of $I(V;T|U)$ at $\mu = 0$. To this end, we notice that by the degeneracy of the chosen pmf,

$$I(V;T|U=u_1) = I(V;T|U=u_2) = 0. \tag{104}$$

Moreover, for any $u \in \mathcal{B}$:

$$I(V;T|U=u) = \mu\nu_0 I^{\mathcal{B}}(V;T|U=u), \tag{105}$$

and thus

$$\frac{\mathrm{d}}{\mathrm{d}\mu} I(V;T|U)\bigg|_{\mu=0} = \nu_0 I^{\mathcal{B}}(V;T|U). \tag{106}$$

B. General Parametrization of Section V

Following similar steps as in the previous subsection, we can conclude that

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}\mu} I(U;Y_1) &= \sum_{u,y_1} \frac{\mathrm{d}P_{UY_1}(u,y_1)}{\mathrm{d}\mu} \log \frac{P_{Y_1|U}(y_1|u)}{P_{Y_1}(y_1)} \\ &\quad + \sum_{u,y_1} P_{UY_1}(u,y_1) \frac{\mathrm{d}}{\mathrm{d}\mu} \log \frac{P_{Y_1|U}(y_1|u)}{P_{Y_1}(y_1)} &(107)\\ &= \sum_{u,y_1} \frac{\mathrm{d}P_{UY_1}(u,y_1)}{\mathrm{d}\mu} \log \frac{P_{Y_1|U}(y_1|u)}{P_{Y_1}(y_1)}, &(108)\end{aligned}$$

and thus when evaluated at $\mu = 0$:

$$\frac{\mathrm{d}}{\mathrm{d}\mu} I(U;Y_1)\Big|_{\mu=0} = \sum_{u,y_1} \frac{\mathrm{d}P_{UY_1}(u,y_1)}{\mathrm{d}\mu} \log \frac{P_{Y_1|U}(y_1|u)}{P_{Y_1}(y_1)}\Big|_{\mu=0} \tag{109}$$

$$\overset{(a)}{=} \sum_{u\in\mathcal{B},y_1} \frac{\mathrm{d}P_{UY_1}(u,y_1)}{\mathrm{d}\mu} \log \frac{P_{Y_1|U}(y_1|u)}{P_{Y_1}(y_1)}\Big|_{\mu=0} \tag{110}$$

$$= \nu_0 \left[I^{\mathcal{B}}(U;Y_1) + D\left(P_{Y_2}^{\mathcal{B}} \middle\| P_{Y_2|X}(\cdot|x_0)\right)\right], \tag{111}$$

where in (a) we used the fact that for $\mu = 0$ the ratio $\frac{P_{Y_1|U}(y_1|u)}{P_{Y_1}(y_1)} = 1$ for $u \in \mathcal{A}$, see (58).

In a similar way, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}\mu} I(U;Y_2)\Big|_{\mu=0} = \nu_0 \left[I^{\mathcal{B}}(U;Y_2) + D\left(P_{Y_2}^{\mathcal{B}} \middle\| P_{Y_2|X}(\cdot|x_0)\right)\right]. \tag{112}$$

We next consider the mutual information $I(U,T;Y_2)$. Following similar steps as before, we have:

$$\frac{\mathrm{d}}{\mathrm{d}\mu} I(U,T;Y_2)$$

$$= \sum_{u,t,y_2} \frac{\mathrm{d}P_{UTY_2}(u,t,y_2)}{\mathrm{d}\mu} \log \frac{P_{Y_2|UT}(y_2|u,t)}{P_{Y_2}(y_2)}$$

$$+ \sum_{u,t,y_2} P_{UTY_2}(u,t,y_2) \frac{\mathrm{d}}{\mathrm{d}\mu} \log \frac{P_{Y_2|UT}(y_2|u,t)}{P_{Y_2}(y_2)} \tag{113}$$

$$= \sum_{u,t,y_2} \frac{\mathrm{d}P_{UTY_2}(u,t,y_2)}{\mathrm{d}\mu} \log \frac{P_{Y_2|UT}(y_2|u,t)}{P_{Y_2}(y_2)}, \tag{114}$$

and when evaluated at $\mu = 0$:

$$\frac{\mathrm{d}}{\mathrm{d}\mu} I(U,T;Y_2)\Big|_{\mu=0}$$

$$= \sum_{u,t,y_2} \frac{\mathrm{d}P_{UTY_2}(u,t,y_2)}{\mathrm{d}\mu} \log \frac{P_{Y_2|UT}(y_2|u,t)}{P_{Y_2}(y_2)}\Big|_{\mu=0} \tag{115}$$

$$\overset{(a)}{=} \sum_{\substack{u\in\mathcal{A},y_2,\\ t\in\mathcal{T}_0(u)}} \frac{\mathrm{d}P_{UTY_2}(u,t,y_2)}{\mathrm{d}\mu} \log \frac{P_{Y_2|UT}(y_2|u,t)}{P_{Y_2}(y_2)}\Big|_{\mu=0}$$

$$+\nu_0 \left[I^{\mathcal{B}}(U,T;Y_2) + D\left(P_{Y_2}^{\mathcal{B}} \middle\| P_{Y_2|X}(\cdot|x_0)\right)\right], \tag{116}$$

$$\overset{(b)}{=} \sum_{\substack{u\in\mathcal{A},y_2\\ t\in\mathcal{T}_0(u)}} \alpha_n P_U^{\mathcal{A}}(u) P_{T|U=u}^{01}(t|u) P_{Y_2|TU=u}^{01}(y_2|t) \cdot \log \frac{P_{Y_2|TU=u}^{01}(y_2|t)}{P_{Y_2|X}(y_2|x_0)}$$

$$+\nu_0 \left[I^{\mathcal{B}}(U,T;Y_2) + D\left(P_{Y_2}^{\mathcal{B}} \middle\| P_{Y_2|X}(\cdot|x_0)\right)\right] \tag{117}$$

$$= \sum_{\substack{u\in\mathcal{A},\\ t\in\mathcal{T}_0(u)}} \alpha_n P_U^{\mathcal{A}}(u) P_{T|U=u}^{01}(t|u) \cdot D(P_{Y_2|T,U=u}^{01}(\cdot|t) \| P_{Y_2|X}(\cdot|x_0))$$

$$+\nu_0 \left[I^{\mathcal{B}}(U,T;Y_2) + D\left(P_{Y_2}^{\mathcal{B}} \middle\| P_{Y_2|X}(\cdot|x_0)\right)\right] \tag{118}$$

where (a) holds by partitioning the (u,t) values and because as explained in (65) for $t \in \mathcal{T}_1(u)$

$$\log \frac{P_{Y_2|UT}(y_2|u,t)}{P_{Y_2}(y_2)}\Big|_{\mu=0} = 0; \tag{119}$$

and (b) holds because for $u \in \mathcal{A}$ and $t \in \mathcal{T}_0(u)$:

$$P_{UTY_2}(u,t,y_2) = (1-\mu) P_U^{\mathcal{A}}(u) \mu\alpha_u P_{T|U=u}^{01}(t) P_{Y_2|TU=u}^{01}(y_2|t), \tag{120}$$

where $P_{T|U=u}^{01}$ and $P_{Y_2|T,U=u}^{01}$ are defined in (68) and (69), and by (70).

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